

Role of Emotional Intelligence in Engineering Leadership

Group - 6

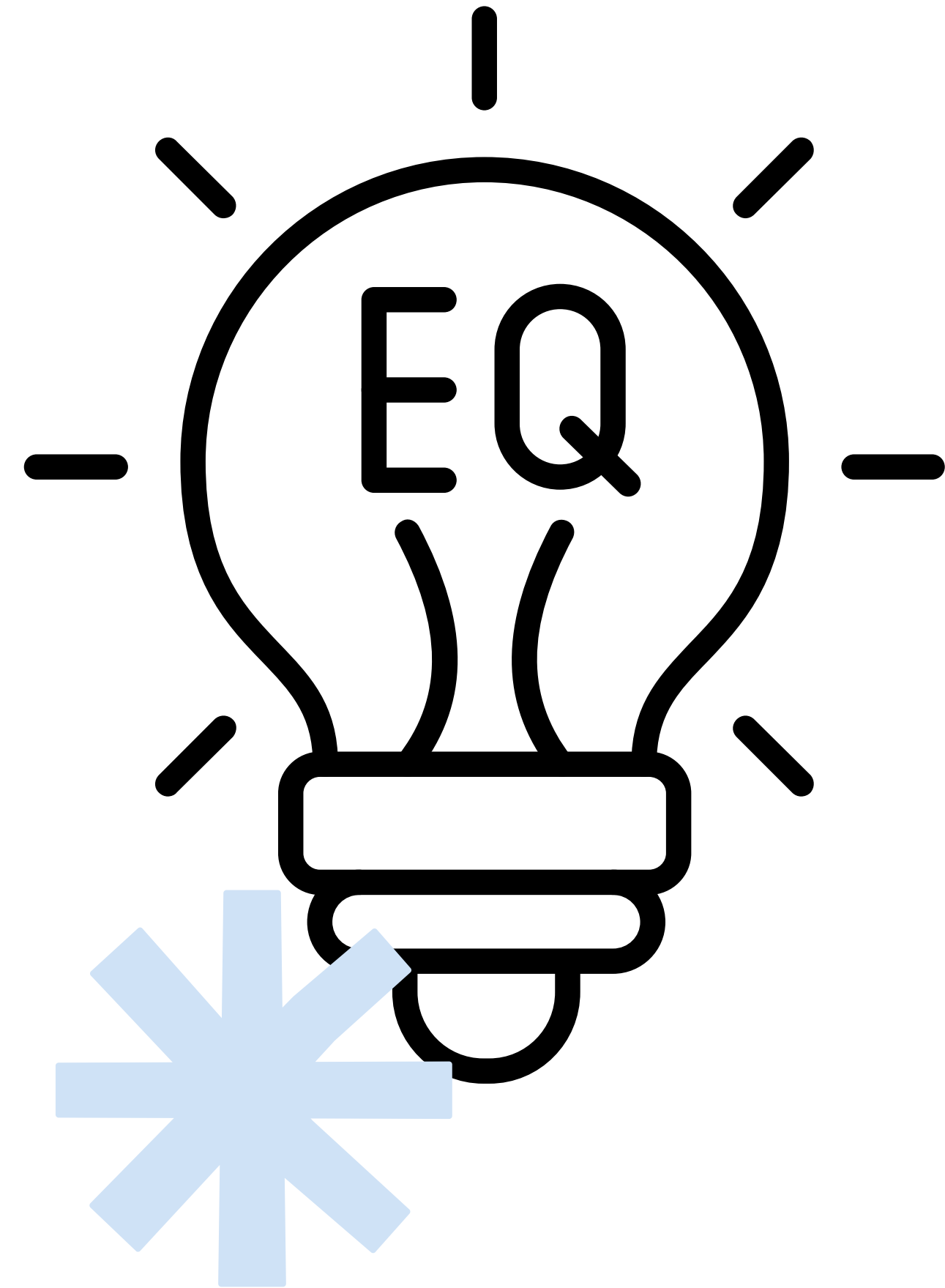


IE PRESENTATION
ROLL NO : 61 ~ 69



Emotional Intelligence (EI)

- **Definition :** Ability to recognize, understand, and manage one's own emotions and those of others.
- **Core Dimensions:**
 - a. **Personal Competence** – Managing self through emotional awareness, self-control, and adaptability.
 - b. **Social Competence** – Managing relationships through empathy, influence, and collaboration.
- **Why It Matters :** Balances technical expertise (IQ) with human skills (EQ), creating leaders who are both capable and compassionate.



Components of Emotional Intelligence

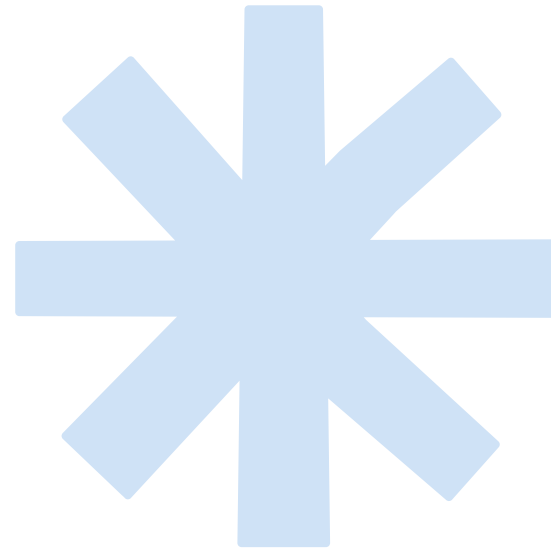
DANIEL GOLEMAN'S MODEL



Components of Emotional Intelligence

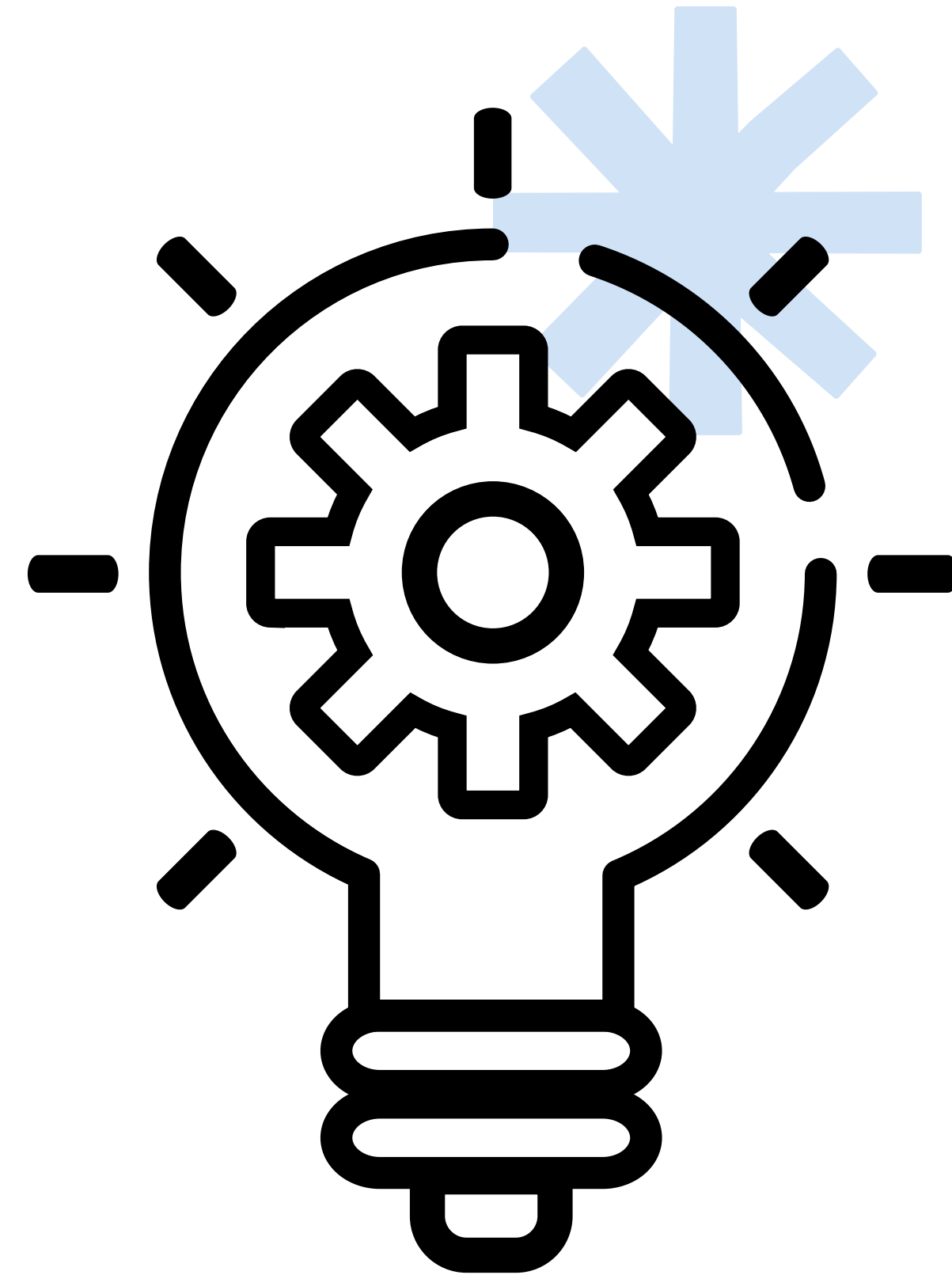
DANIEL GOLEMAN'S MODEL

1. **Self-Awareness** - Recognizing one's emotions and their impact (Emotional Self-Awareness).
2. **Self-Management** -
 - a. Maintaining control under stress (Emotional Balance).
 - b. Adapting to change (Adaptability).
 - c. Staying motivated and optimistic (Achievement Orientation, Positive Outlook).
3. **Social Awareness** -
 - a. Understanding others' emotions (Empathy).
 - b. Reading group dynamics and culture (Organizational Awareness).
4. **Relationship Management** -
 - a. Guiding and influencing others (Influence).
 - b. Handling disagreements constructively (Conflict Management).
 - c. Building strong teams (Teamwork).
 - d. Inspiring and developing people (Inspirational Leadership, Coaching & Mentoring).



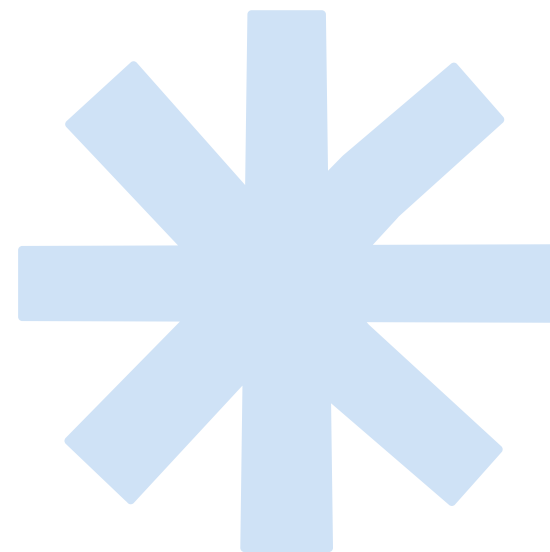
Engineering Leadership

- **Definition :** The ability of engineers to lead teams, manage projects, and align technical expertise with organizational goals.
- **Key Roles :**
 - a. **Technical Guidance** – Providing expertise and ensuring quality solutions.
 - b. **People Management** – Motivating and supporting team members.
 - c. **Decision-Making** – Balancing technical, business, and human factors.
 - d. **Innovation & Problem-Solving** – Driving creative solutions to complex challenges.
- **Why It Matters :** Engineering leadership ensures projects are not only technically sound but also collaborative, efficient, and sustainable.



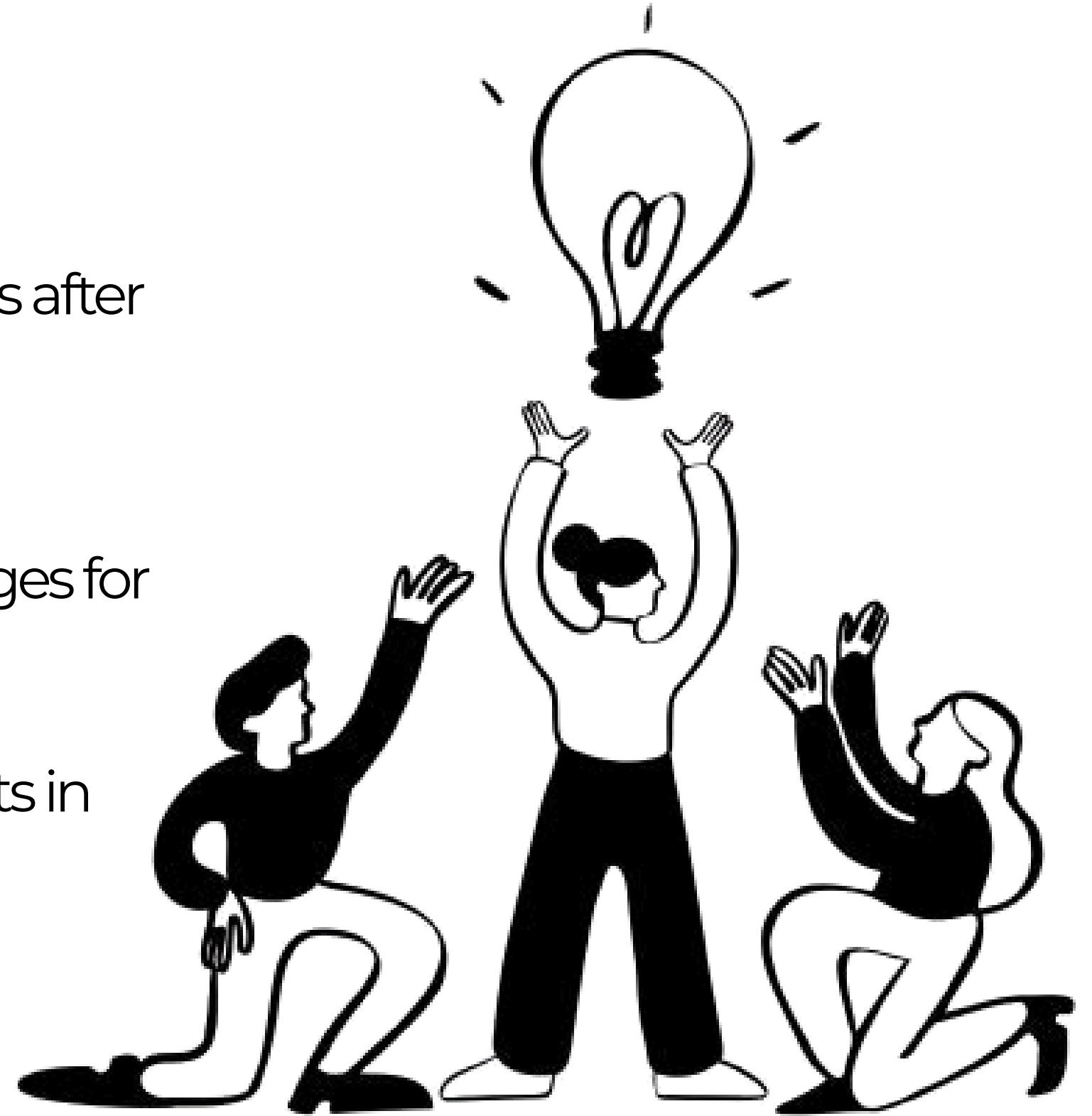
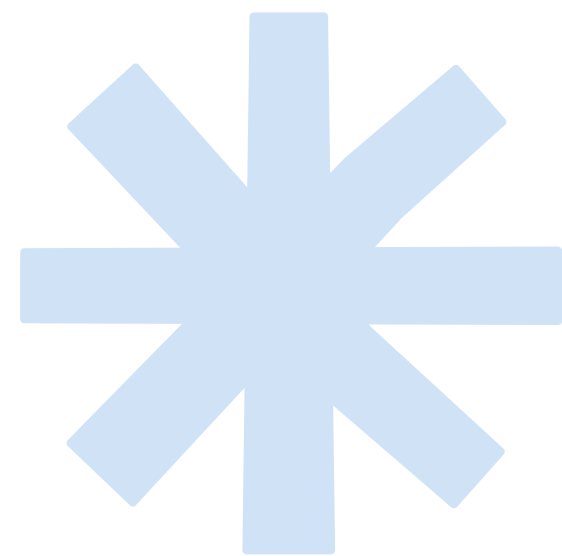
Importance Of EL

- Engineers today **lead diverse, cross-functional teams.**
- Leaders with **Emotional Intelligence (EI)** can:
 - a. Foster **collaboration** across technical and non-technical roles.
 - b. Resolve conflicts in a **constructive and fair** manner.
 - c. **Motivate and inspire** technical teams to achieve goals.
 - d. Stay composed and guide others under **high-pressure conditions.**

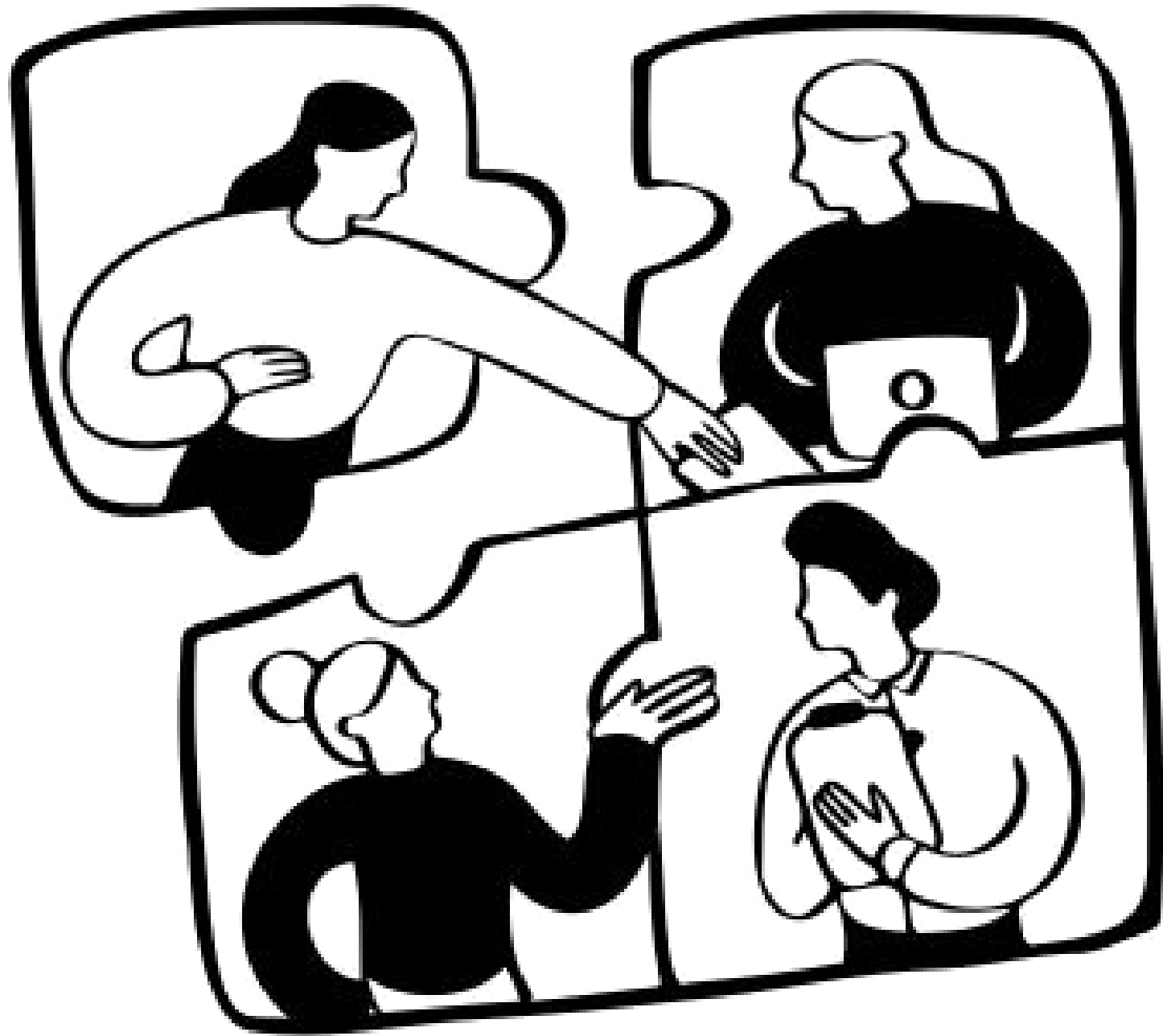
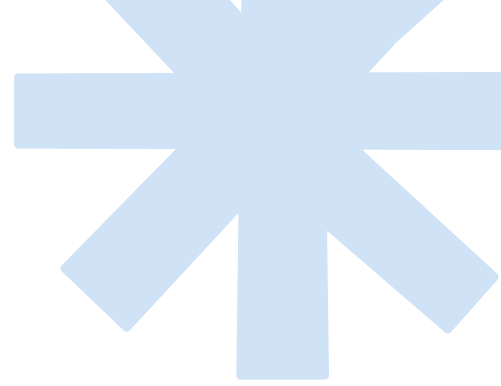


Role of EI in Leadership

- **Decision-Making**
 - Balances logic with empathy.
 - Considers both technical facts and human factors.
- **Example :** An engineering manager adjusting deadlines after understanding team stress levels.
- **Communication**
 - Translates complex technical ideas into clear messages for non-technical stakeholders.
 - Builds trust through transparency and clarity.
- **Example :** A project lead explaining system risks to clients in simple terms.



Role of EI in Leadership



- **Conflict Resolution**
 - Uses empathy and fairness to address disagreements constructively.
 - Focuses on solutions instead of blame.
- **Example :** Mediating between design and development teams with respect to both sides' concerns.
- **Innovation & Creativity**
 - Encourages open sharing of ideas without fear of judgment.
 - Creates a safe, collaborative environment for problem-solving.
- **Example :** Leaders at Google promoting “psychological safety” so engineers can share bold ideas.

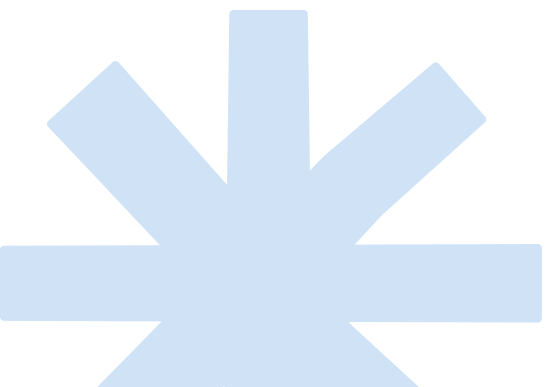
Eg - 1 : System Failure Scenario

- Project manager stays calm during a sudden system failure.
- Uses EI to reduce panic and keep the team focused.
- Guides team toward quick, logical problem-solving.

1. In this example, a project manager faces a sudden system failure.
2. Instead of letting panic spread, the leader stays calm, reassures the team, and provides clear instructions.
3. By managing emotions effectively, the manager prevents chaos and enables the team to focus on finding a solution quickly.
4. This shows how EI supports calmness and clarity in high pressure situations.

Real Life Exmample in Reference :

NASA Apollo 13 Mission (1970)



Eg -2 : Burnout Recognition Scenario

- Engineering leader notices signs of team fatigue and burnout.
- Uses EI to empathize with team members' stress.
- Adjusts workload and provides support to restore motivation.

1. In this case, an engineering leader notices that the team is showing signs of burnout - missed deadlines, lack of energy, and disengagement.
2. Instead of pushing harder, the leader empathizes, listens, and adjusts workload distribution.
3. This not only prevents further stress but also rebuilds trust and motivation.
4. It highlights how EI can help leaders sustain long-term team performance.

Real Life Exmample in Reference :

Google's Project Aristotle



Benefits & Development

BENEFITS OF EI

- Builds stronger **team cohesion**.
- Boosts **productivity and innovation**.
- Reduces **stress and employee turnover**.
- Strengthens **client and stakeholder relationships**.

DEVELOPING EI

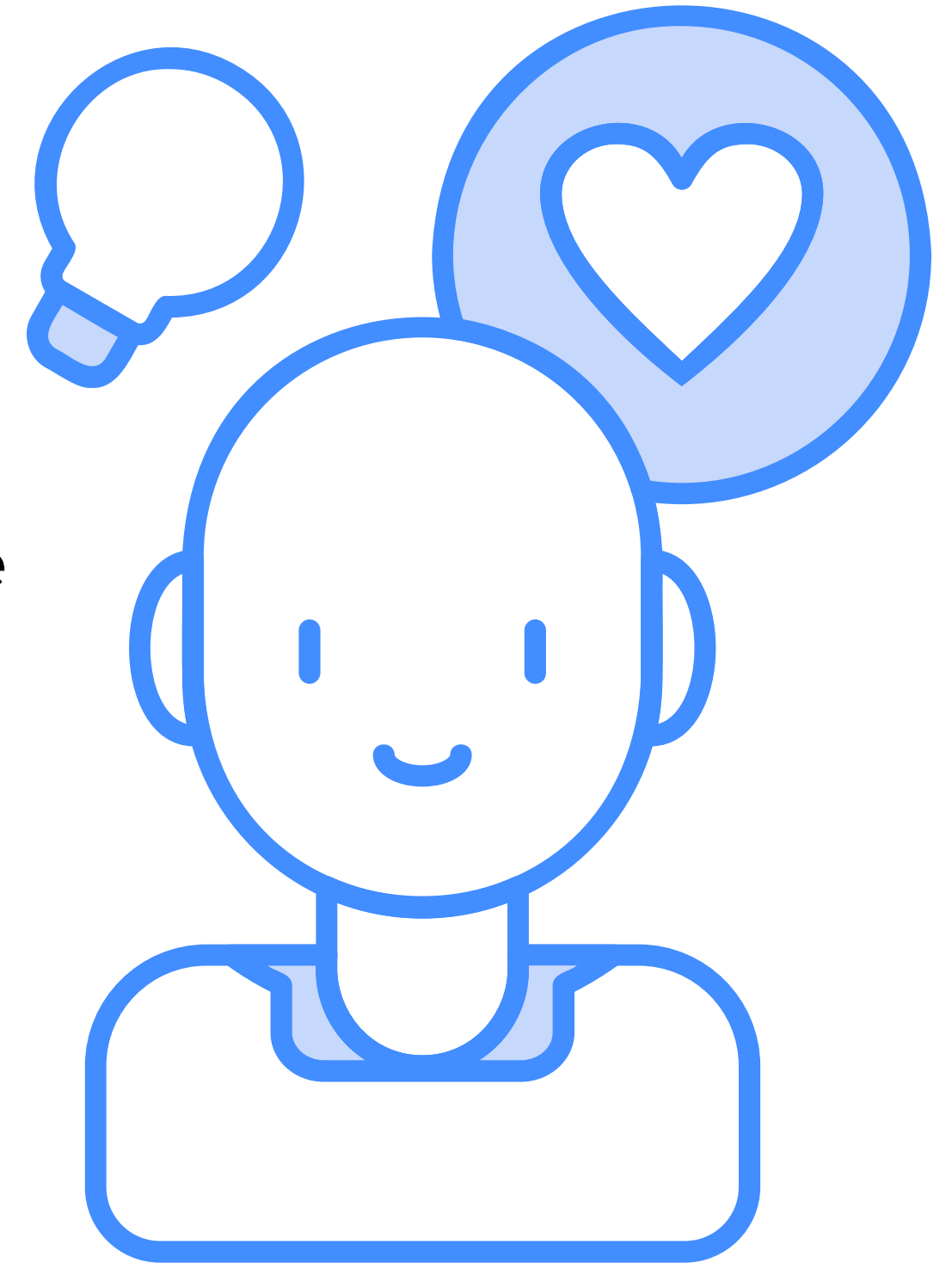
- Practice **self-reflection** to understand emotions.
- Improve **listening and empathy skills**.
- Give and receive **constructive feedback**.
- Build **cultural and social awareness**.
- Use **stress-management practices**

Conculsion

- EI is as important as **technical skills** for engineering leaders.
- Leaders with EI build **trust, motivation, and long-term success.**
- *“An engineer with EI is not just a problem-solver, but a people-leader.”*

Emotional Intelligence is just as critical as technical skills in engineering leadership. Leaders with strong EI foster trust, motivate their teams, and ensure long-term success.

An engineer with EI goes beyond solving technical problems ; they also lead people effectively, which is the true mark of leadership



END

Thank You

The Dream is to Keep Surprising yourself

-Tom Hiddleston